

Can Bacteria Make Carbon Capture Faster?

A Proposed Field Study of Microbially Accelerated Enhanced Rock Weathering

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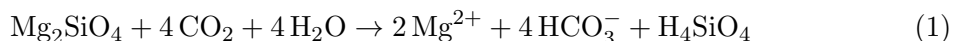
Introduction

Mean temperatures have risen approximately 1.2°C, and IPCC AR6 recommendations aimed at limiting warming to 1.5°C require more than 300 Gt cumulative CO₂ removal by 2100, with annual rates reaching 7–9 Gt CO₂ by mid-century (IPCC, 2021; Smith et al., 2024). Current deployment is around 0.0013 Gt CO₂ yr⁻¹, leaving a gap of approximately four orders of magnitude between current and target states (Smith et al., 2024).

Enhanced rock weathering (ERW) quickens the slow carbon cycle by spreading crushed silicate rock onto soils. Rainwater reacts with the rock, triggering chemical weathering that converts atmospheric CO₂ into dissolved bicarbonate ions (HCO₃⁻), carrying the carbon through soil to the ocean, where it is sequestered for millennia (Beerling et al., 2020). Beerling et al. (2020) estimated that basalt-ERW could remove 0.5–2 Gt CO₂ yr⁻¹, and a subsequent four-year study confirmed this by applying 50 t ha⁻¹ yr⁻¹ of crushed basalt to maize-soybean rotations and recording 10.5 ± 3.8 t CO₂ ha⁻¹ of cumulative removal alongside crop yield increases of 12–16% (Beerling et al., 2024).

The carbon capture happens when basalt weathers through the dissolution of olivine ((Mg,Fe)₂SiO₄) and pyroxene (MgSiO₃). Olivine is a nesosilicate consisting of isolated SiO₄ tetrahedra bonded through ionic Mg–O and Fe–O linkages, with no Si–O–Si covalent bridges connecting adjacent tetrahedra (Oelkers et al., 2018). Pyroxene is a single-chain inosilicate, meaning its SiO₄ tetrahedra are linked in chains by sharing two oxygens per tetrahedron with adjacent units, so while pyroxene has Si–O–Si bridges, they lack the three-dimensional network present in more resistant minerals (Oelkers & Schott, 2001). Quartz shares all four oxygens per tetrahedron, forming a fully polymerized framework with no ionic metal-oxygen bonds for protons to attack. During dissolution, water molecules and hydrogen ions attack more bonds simultaneously where fewer Si–O–Si bridges exist, making dissolution faster, a relationship the Goldich dissolution series captures empirically by placing olivine as the fastest-weathering silicate and quartz as the slowest (Goldich, 1938; Oelkers et al., 2018; Oelkers & Schott, 2001).

When olivine dissolves, the reaction consumes four CO₂ molecules:



fixing carbon as bicarbonate in solution (Beerling et al., 2020). However, natural field weathering rates are 2–4 orders of magnitude slower than laboratory experiments, because secondary-mineral precipitation protects reactive surfaces, proton supply at circumneutral soil pH is limited, and fluid residence times are slower (White & Brantley, 2003).

Bacillus mucilaginosus polysaccharides adsorb onto mineral surfaces and concentrate organic acids, generating a low-pH microenvironment that quickens dissolution while adsorbing dissolved SiO₂ away from the mineral surface, shifting the dissolution equilibrium toward cation release (Liu et al., 2006). Separately, *B. mucilaginosus* produces carbonic anhydrase

(CA), an enzyme that catalyzes $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}^+$ at rates 10^6 – 10^7 times faster than the uncatalyzed reaction (Smith & Ferry, 2000). Zhang et al. (2011) demonstrated that when CA and *B. mucilaginosus* act together on carbonate minerals, Ca^{2+} release reaches concentrations 3–6 times higher than either CA or bacteria alone, with pH rising from 7.1 to 8.4.

Polysaccharide-mediated product removal pulls the dissolution equilibrium forward, which in turn maintains the low-product-concentration condition that keeps CA-driven CO_2 hydration from slowing as HCO_3^- accumulates, creating a positive feedback loop that promotes acceleration (Liu et al., 2006; Zhang et al., 2011). Whether this microbial acceleration translates to basalt agricultural soil, and whether bicarbonate production reaches measurable concentrations as soil dissolved inorganic carbon (SDIC), has not been tested under field conditions. Wang et al. (2013) showed that SDIC cycles through the soil profile faster than solid inorganic carbon, making it the most responsive measurable variable if *B. mucilaginosus* increases basalt mineral weathering.

We therefore hypothesize that injecting *B. mucilaginosus* into basalt-amended soil will increase SDIC concentration in the soil profile, because the bacterium’s coupled polysaccharide-CA mechanism pulls weathering reactions forward and captures the released CO_2 as stable bicarbonate. We predict that this increase will be detectable at depths of 60 cm and 120 cm within two growing seasons, with the bacteria-basalt treatment producing higher Ca^{2+} and Mg^{2+} cation concentrations in collected soil water than the basalt-only control.

Research Design

The proposed study takes place at the University of Illinois Energy Farm in Champaign County, Illinois (40°30’N, 88°11’W), the same site used by Beerling et al. (2024) for the first validated Corn Belt ERW field trial. The site has dark organic-rich soils suited to ERW because nitrogen fertilization has acidified them, creating demand for the pH-raising effect that basalt amendment provides (Beerling et al., 2020). Annual precipitation of around 1,000 mm is sufficient to move dissolved weathering products through the soil.

The study uses a 2×2 factorial design, testing four treatment groups arranged to evaluate each factor independently and in combination: control soil, basalt-only amendment, *B. mucilaginosus* inoculant only, and basalt combined with *B. mucilaginosus*. Each treatment is replicated across four randomized blocks under a maize-soybean rotation for two consecutive growing seasons.

The basalt material is the same Blue Ridge meta-basalt used by Beerling et al. (2024), applied at 50 t ha^{-1} and ground to a median particle size of $d_{80} = 267 \text{ }\mu\text{m}$, meaning 80% of particles are smaller than 267 micrometers, maximizing reactive surface area for dissolution. Basalt is an aphanitic igneous rock, meaning its mineral grains are too fine to see with the naked eye, a texture resulting from rapid cooling of mafic magma that produces a dense structure of olivine and pyroxene crystals whose high surface-area-to-volume ratio, when crushed, makes it substantially more reactive (Beerling et al., 2020). The bacterial inoculant is prepared from a *B. mucilaginosus* liquid culture and applied at the time of basalt spreading each fall, so bacteria colonize the mineral surfaces before the growing season starts.

To measure changes in dissolved carbon moving through the soil, lysimeters (porous ceramic cup samplers used to collect soil pore water) are installed at 60 cm and 120 cm depth in

each plot. Soil water is collected every two weeks during the growing season and analyzed for SDIC, dissolved Ca^{2+} and Mg^{2+} , and dissolved Si. Cation mass balance, following the methodology of Reershemius et al. (2023), converts these soil water concentrations into an estimate of weathering-derived CO_2 removal. At the end of each growing season, soil from each plot is also analyzed to cross-check lysimeter-derived carbon removal estimates against changes in the soil’s solid phase.

Preliminary Results

The preliminary evidence for the proposed mechanism comes from Zhang et al. (2011), who tested the effect of bacterial CA enzyme on mineral dissolution in a laboratory system by growing *B. mucilaginosus* in limestone-containing medium under four conditions: pure water as a control, CA enzyme alone, bacteria alone, and bacteria combined with CA. As shown in Figure 1, the bacteria-CA treatment produced the highest Ca^{2+} release after ten hours, showing synergistic acceleration beyond what either mechanism achieves independently.

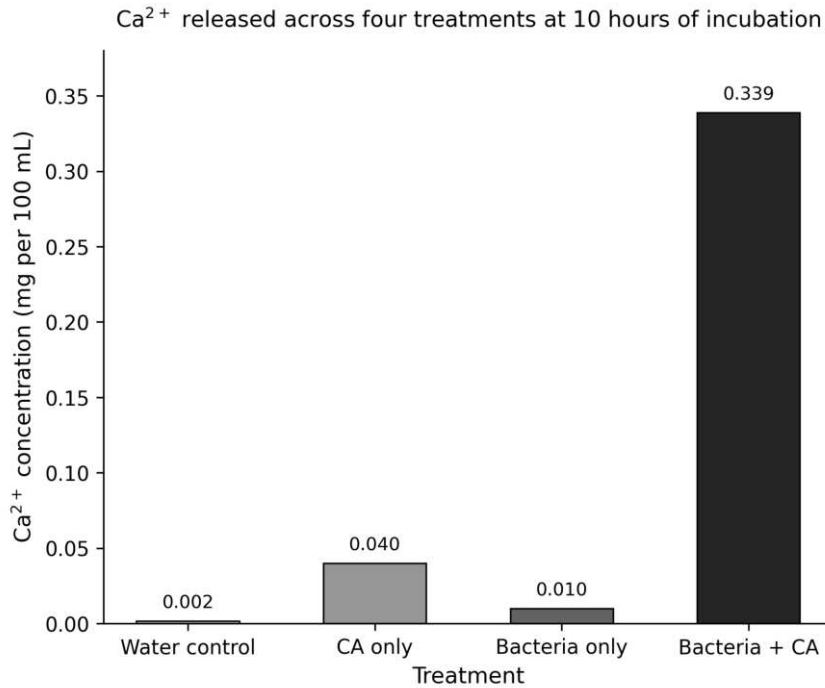


Figure 1: **Ca²⁺ released across four treatments at 10 hours of incubation**, recreated from Zhang et al. (2011) Figure 4. The bacteria-CA bar is the highest at 0.339 mg per 100 mL, as reported in the paper text. The CA-only bar reaches approximately 0.040 mg per 100 mL, roughly four times the control value. The bacteria-only bar reaches approximately 0.010 mg per 100 mL, and the water control sits at approximately 0.0018 mg per 100 mL. The bacteria-CA treatment reached approximately three times the CA-only concentration and six times the bacteria-only concentration, showing that the two mechanisms together produce acceleration that neither achieves independently. The underlying data and Python script used to generate this figure are available at https://drive.google.com/drive/folders/1CB0cM_AHWgjIyipKMivnuuAKN4DBPufd?usp=sharing.

The pH increase from 7.1 to 8.4 in the bacteria-CA treatment shows how the interaction of four Earth surface components produces a chemical outcome that none could achieve alone. The CA enzyme produced by *B. mucilaginosus* accelerates CO₂ hydration roughly 10⁶ times faster than the uncatalyzed reaction, generating H⁺ that attacks the mineral surface and releases Ca²⁺ into the surrounding water, while the accumulation of HCO₃⁻ drives pH past 8.3 and shifts the carbonate equilibrium away from soluble carbonic acid toward stable bicarbonate. Without the enzyme there is no acceleration, without the mineral there are no cations to balance the charge, and without water as the solvent the entire reaction chain cannot proceed (Zhang et al., 2011; Smith & Ferry, 2000). Although Zhang et al. (2011) used limestone rather than silicate minerals, Liu et al. (2006) confirmed that *B. mucilaginosus* produces the same polysaccharide-mediated acid attack and product-removal mechanism on silicate minerals, and Buss et al. (2025) showed that fresh basalt amendment raises dissolved inorganic carbon concentrations in temperate cropland soils, providing evidence for extending these findings to the basalt-amended field setting.

Interpretation

If the basalt-bacteria treatment produces substantially higher SDIC concentrations and Ca²⁺ and Mg²⁺ cation fluxes at 60 cm and 120 cm depth compared to the basalt-only control across two growing seasons, the results would support the hypothesis that *B. mucilaginosus* accelerates basalt weathering in agricultural soil through its coupled polysaccharide and carbonic anhydrase mechanisms. Wang et al. (2013) measured SDIC residence times in soil profiles and found that SDIC ages are orders of magnitude younger than solid soil inorganic carbon, with SDIC accumulation rates in the upper meter reaching 7.58–29.54 g C m⁻² yr⁻¹ compared to 1.34–5.33 g C m⁻² yr⁻¹ for solid inorganic carbon in the same layers. SDIC dissolves into percolating soil water and moves downward through the soil profile by advection and diffusion, meaning water physically carries dissolved bicarbonate through pore spaces between soil particles and along macropores until it accumulates in deeper horizons or reaches groundwater, a transport process that Kindler et al. (2011) identified as a crucial component of net ecosystem carbon balance in temperate agricultural soils. Detection of elevated SDIC at 120 cm depth in the bacteria-basalt treatment would indicate that bacterially enhanced dissolution at the mineral surface has produced enough additional bicarbonate to be measurable after transport through more than a meter of soil.

Three alternative explanations need to be ruled out before elevated SDIC can be attributed to bacterial weathering acceleration. The first is that higher SDIC in the bacteria-basalt treatment represents microbial biomass decomposition releasing dissolved organic carbon that the analysis misidentifies as inorganic carbon. Acidifying soil pore water samples before infrared gas analysis volatilizes all inorganic carbonate as CO₂, leaving only organic carbon for a second measurement, and the difference gives the true SDIC fraction, an approach that Reershemius et al. (2023) validated and Wang et al. (2013) applied to distinguish SDIC from total soil inorganic carbon. The second is that Ni, Cr, and Cu released from basalt dissolution accumulate in the soil to concentrations toxic to crops and soil organisms, meaning carbon signals reflect altered soil biology rather than mineral weathering acceleration. Levy et al. (2024) recommended monitoring Ni and Cr in soil pore water, while Dupla et al. (2023) modeled that at application rates of 40 t ha⁻¹ yr⁻¹, Cu thresholds could be exceeded, making pore water metal concentrations a necessary covariate. The third is that spatial variation in soil moisture between blocks creates SDIC differences independent of treatment, because wetter plots move more dissolved carbon to depth. Kindler et al. (2011) showed that dissolved carbon leaching varies by a factor of

four across sites due to differences in drainage intensity, meaning that soil moisture sensors and treating moisture as a statistical covariate are necessary rather than optional.

If *B. mucilaginosus* accelerates CO₂ hydration faster than olivine and pyroxene dissolution can supply Ca²⁺ and Mg²⁺ to balance the charge, solution pH rises above 8.3, shifting the carbonate equilibrium toward carbonate ion (CO₃²⁻) rather than bicarbonate, promoting CaCO₃ precipitation in the soil rather than transport of dissolved carbon. Hartmann et al. (2013) estimated that secondary carbonate precipitation imposes a roughly 50% efficiency penalty on ERW carbon accounting because each mole of CaCO₃ that re-precipitates releases one mole of CO₂ back to the soil atmosphere. The enzymatic mechanism that accelerates carbon capture under one set of conditions therefore becomes a carbon source under another, depending on whether mineral cation supply keeps pace with CO₂ hydration. Monitoring the molar ratio of dissolved Ca²⁺ and Mg²⁺ to SDIC at both depths would reveal which regime dominates the system.

Broader Impacts

This research advances scientific understanding of microbially accelerated silicate weathering, a mechanism studied in laboratory settings but never tested under field conditions in a basalt-amended agricultural system. By generating the first field-scale dataset on how *Bacillus mucilaginosus* affects soil dissolved inorganic carbon flux, cation release rates, and weathering-derived CO₂ removal in Corn Belt soils, the proposed study fills a gap that currently prevents the research community from knowing whether microbial injections can meaningfully improve ERW efficiency.

Corn Belt farmers already apply lime to counteract soil acidification from nitrogen fertilization, and basalt amendment serves the same agronomic function while also removing CO₂. If *B. mucilaginosus* inoculation increases the carbon removal efficiency of that amendment without adding cost or complexity, farmers gain a climate benefit from a practice they are already motivated to adopt for yield reasons. Beerling et al. (2024) showed 12–16% yield increases from basalt alone, and microbially driven amplification of that effect would strengthen the economic case for ERW adoption across the Corn Belt.

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